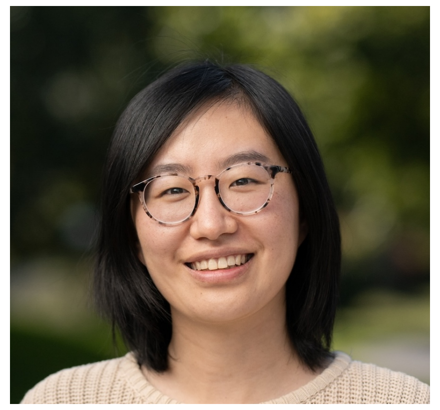
THE UNIVERSITY *of* EDINBURGH

Toward a psychologically grounded theory of collective innovation

Melbourne Lab Talk



Bonan Zhao
Edinburgh,
Informatics

Natalia Vélez
Princeton

Chris Lucas
Edinburgh, Informatics

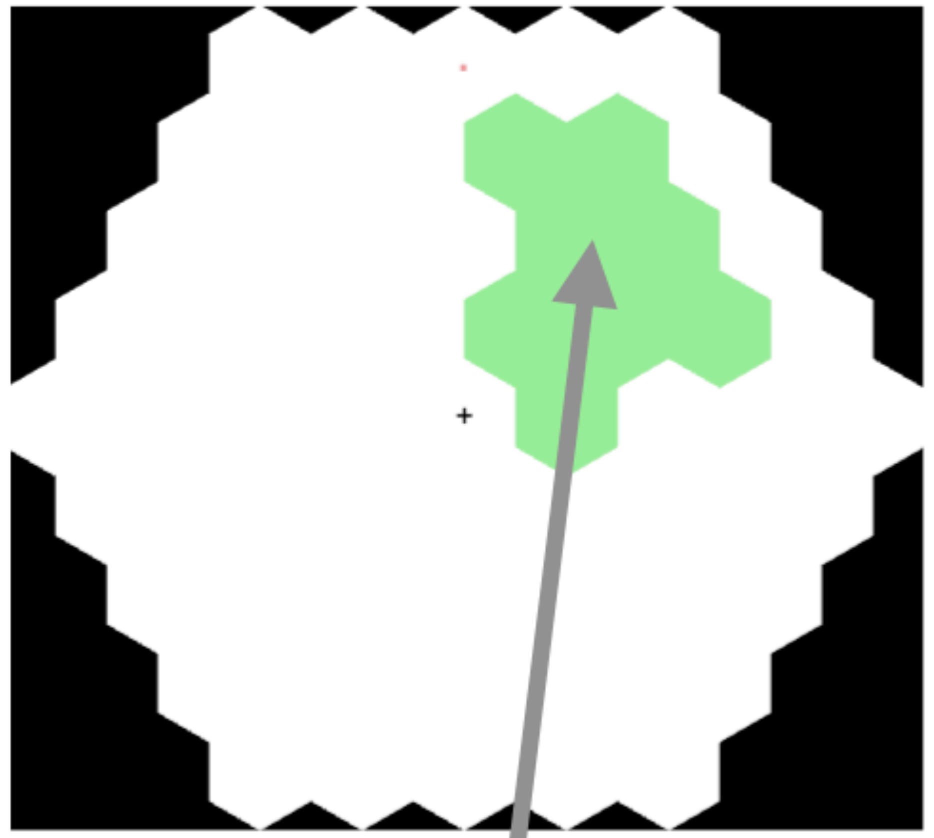
Neil Bramley
Edinburgh, Psychology

Neil Bramley, 6th March, 2026

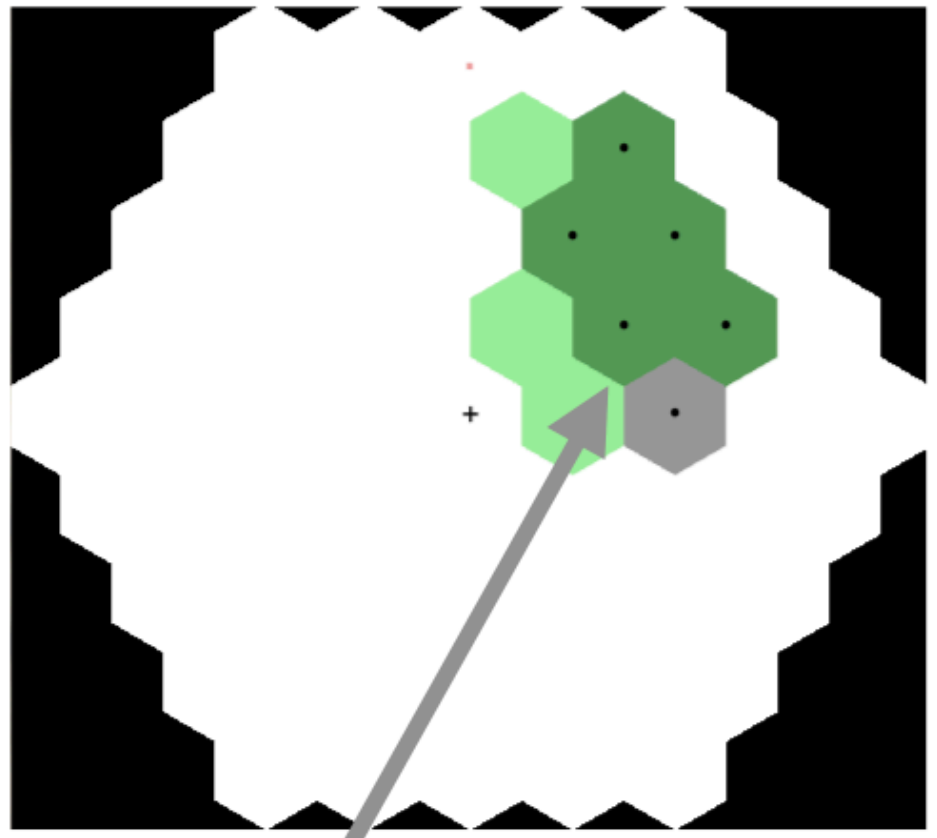
A procedural puzzle task

https://eco.ppls.ed.ac.uk/~nbramley/hexcraft/demo_nocache.html

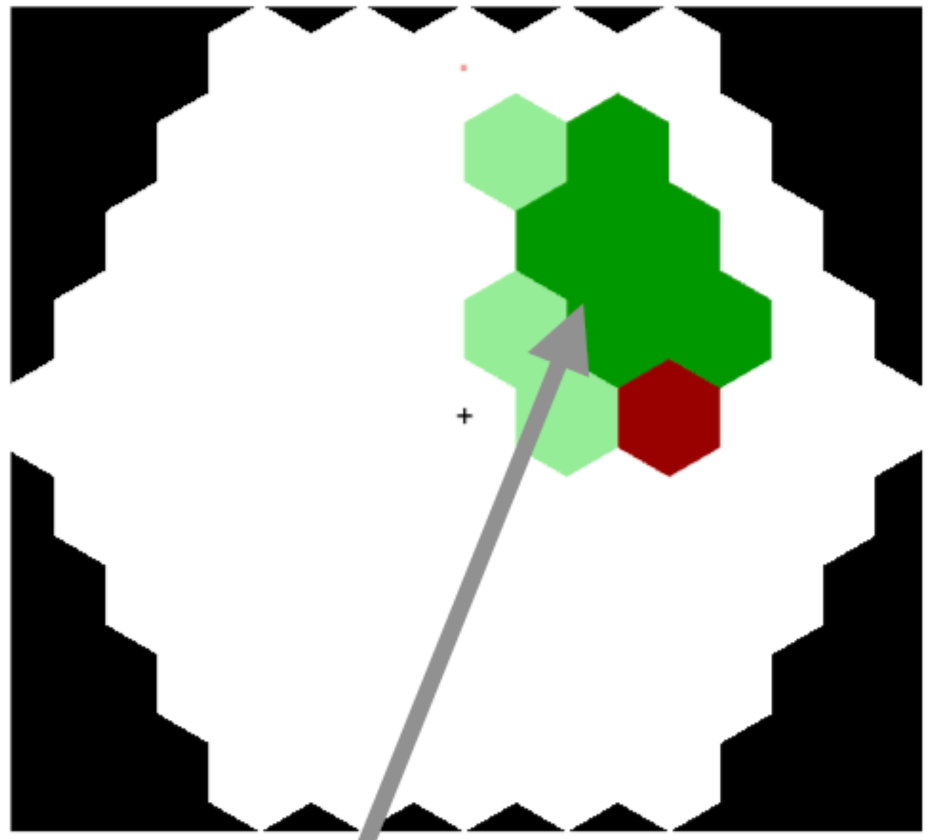
Example puzzles



Target pattern shown in green

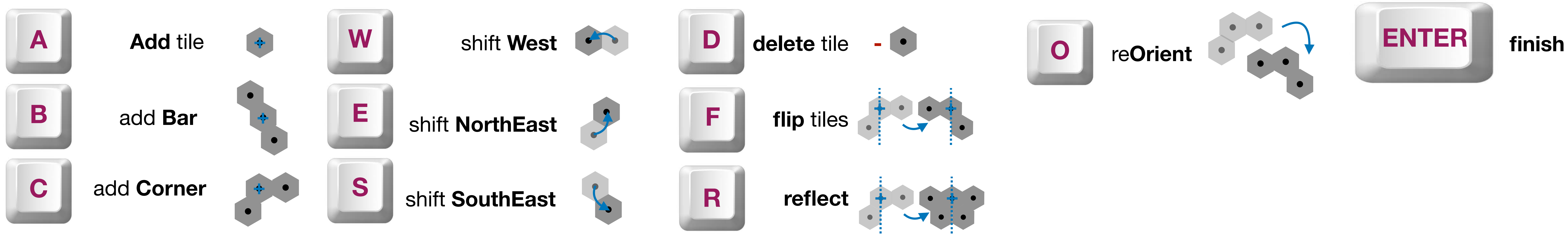


Solution-in-progress shown in transparent grey

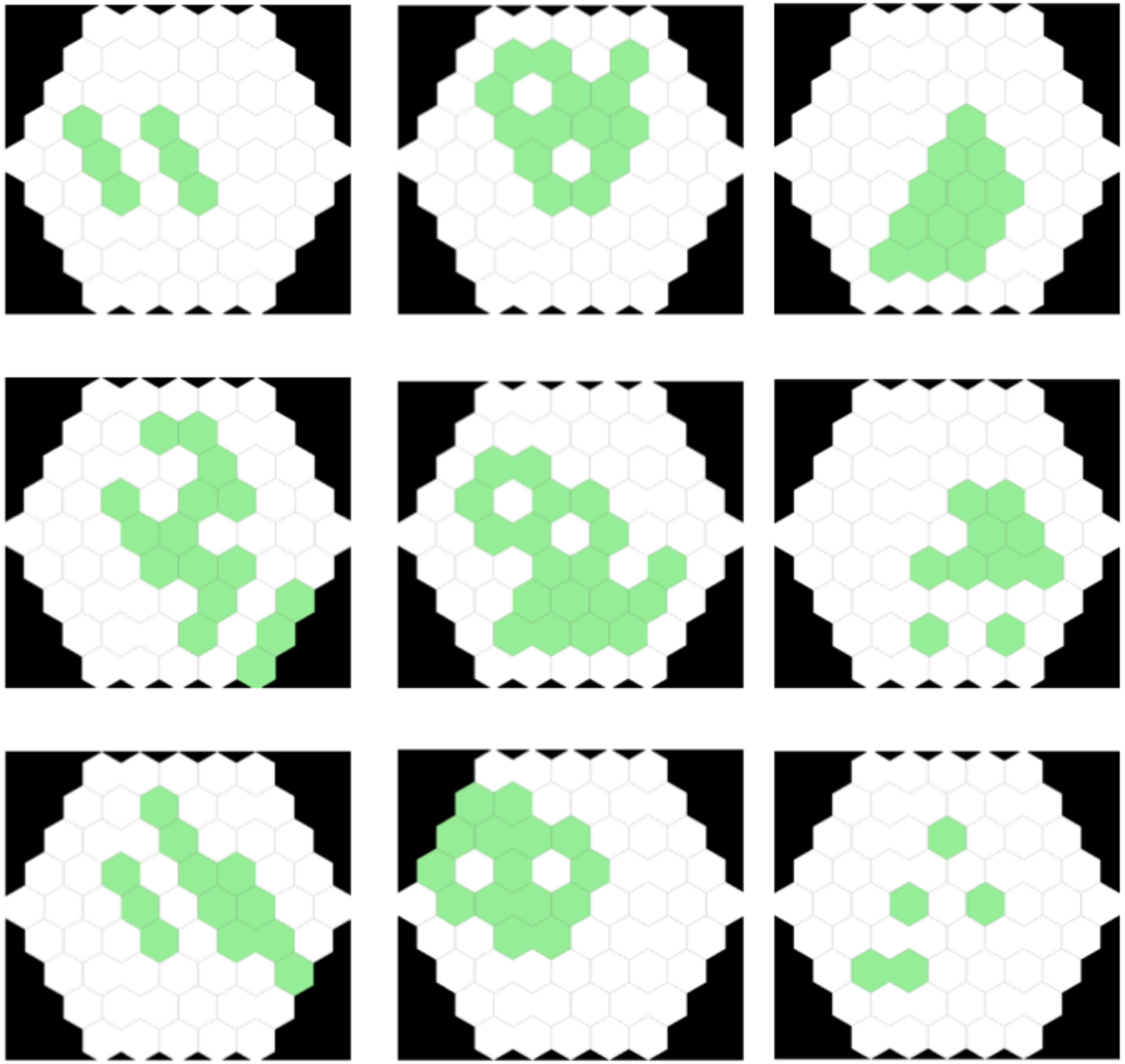


(Incorrect) solution locked in. Matching tiles shown in **dark green**, non-matching tiles in **red**, missing tiles in **light green**

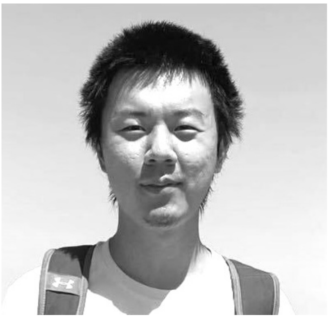
Example action space



Recognisable repeated structures



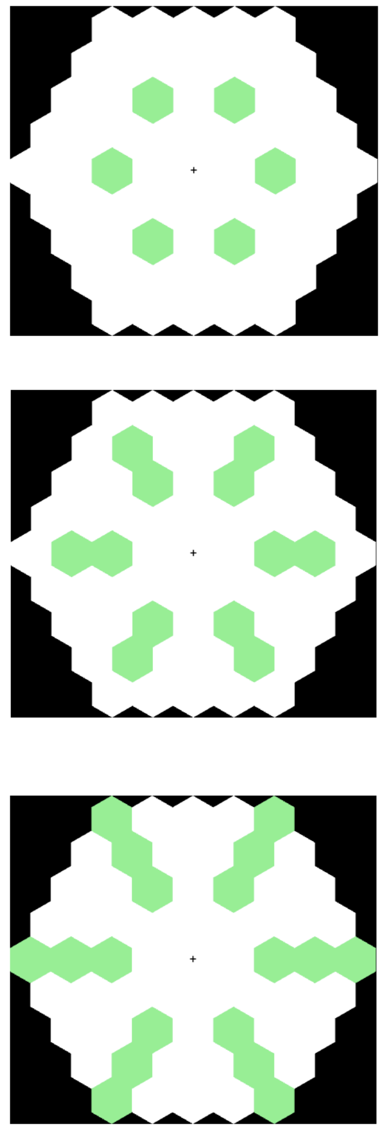
A pilot experiment



Haozhe Sun

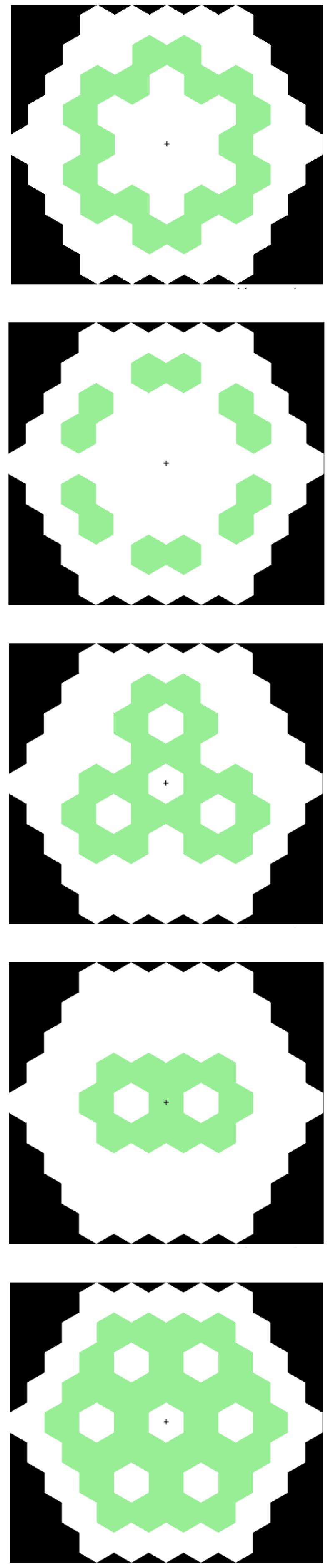
- Before moving to group setting, can individuals form and generalise useful abstractions when solving these sorts of puzzles?
- Can we infer their abstractions by doing library induction on their sequence data?
- Pilot with N=80 Prolific workers
- Shared tutorial, introducing each available action
- 3 “training puzzles” unique to condition with different common sequences embedded, then 5 “test puzzles” shared by both conditions involving one or both sequences
- Had to make at least 5 attempts at each (or succeed), timeout at 30 actions per attempt, bonuses for successful completions

“Reflect” Condition Training



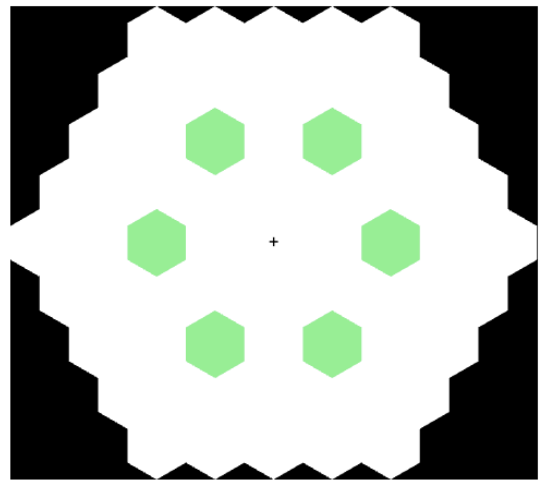
“Flower” Condition Training

Test problems



Pilot experiment

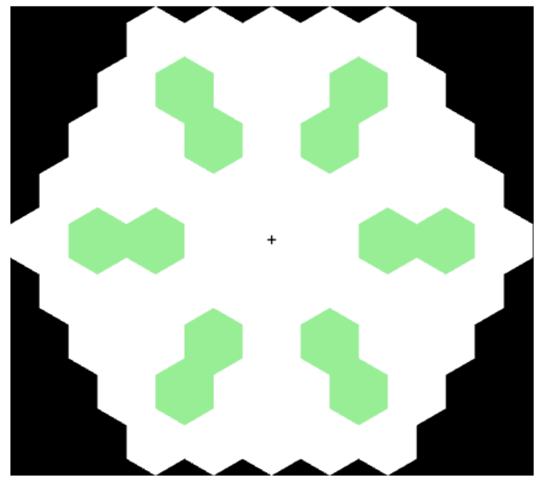
"Reflection" training problems



awwroror

Add**W**est**W**est...

...**R**eflect**O**rient**R**eflect**O**rient**R**eflect



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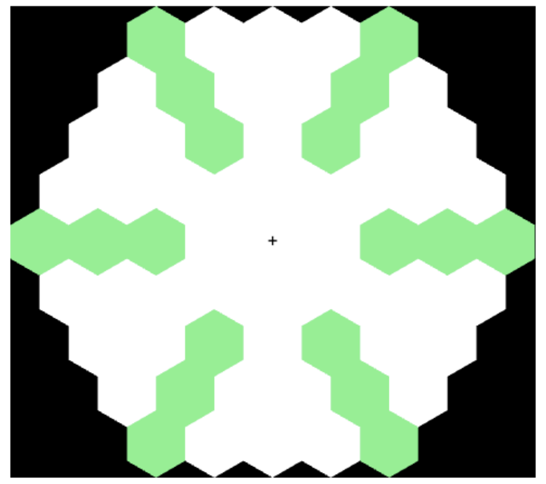
Add**W**est**A**dd**W**est**W**est...

...**R**eflect**O**rient**R**eflect**O**rient**R**eflect

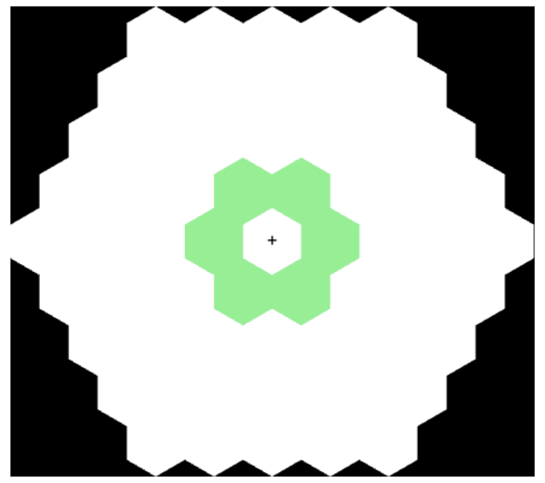


awawawwroror

Add**W**est**A**dd**W**est**A**dd**W**est**W**est...

...**R**eflect**O**rient**R**eflect**O**rient**R**eflect

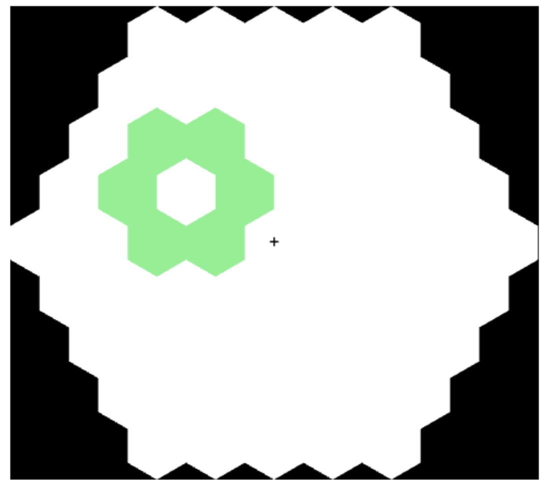
"Flower" training problems



cbdr

Corner**B**ar**D**elete**R**eflect

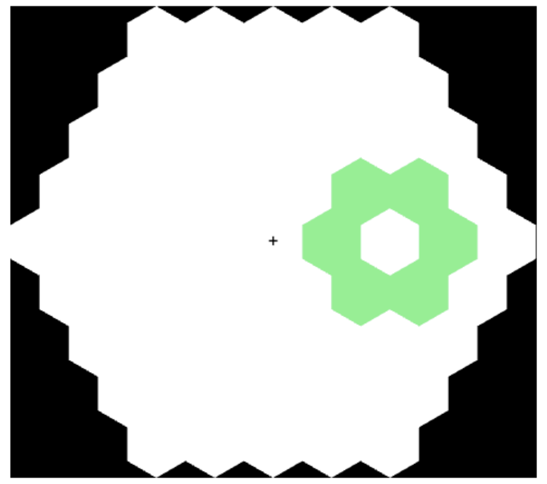
(**B**ar**O**rient**B**ar**O**rient**B**ar**D**elete)



cbdrwwe

Corner**B**ar**D**elete**R**eflect...

...**W**est**W**est(north)**E**ast

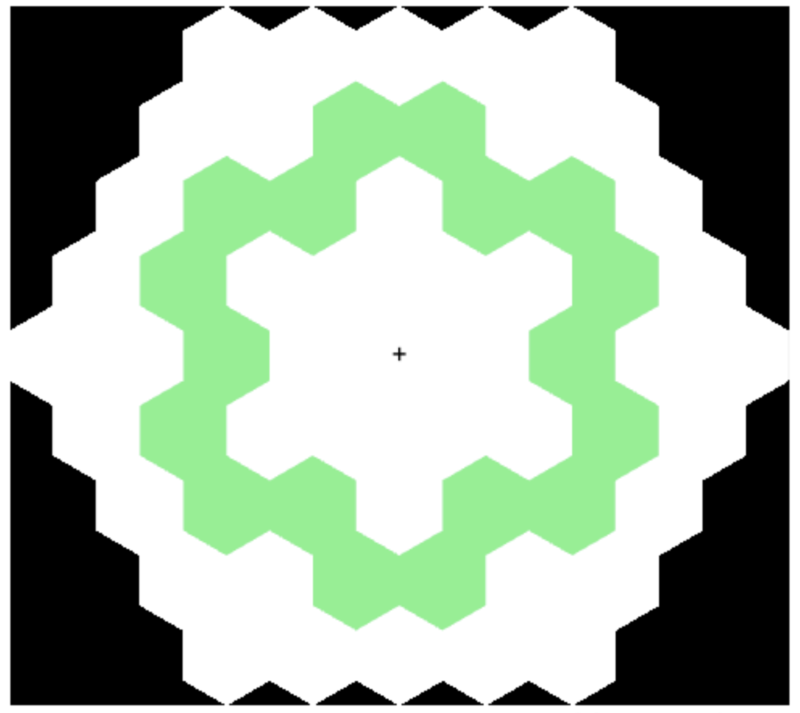


cbdrwwf

Corner**B**ar**D**elete**R**eflect...

...**W**est**W**est**F**lip

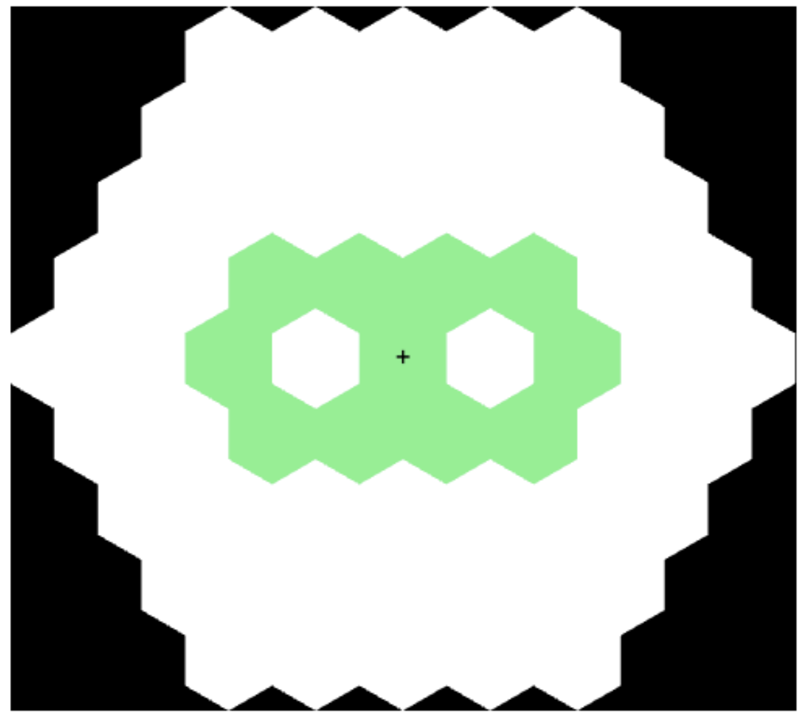
Shared Test Problems



cssroror

Corner**S**outh**S**outh...

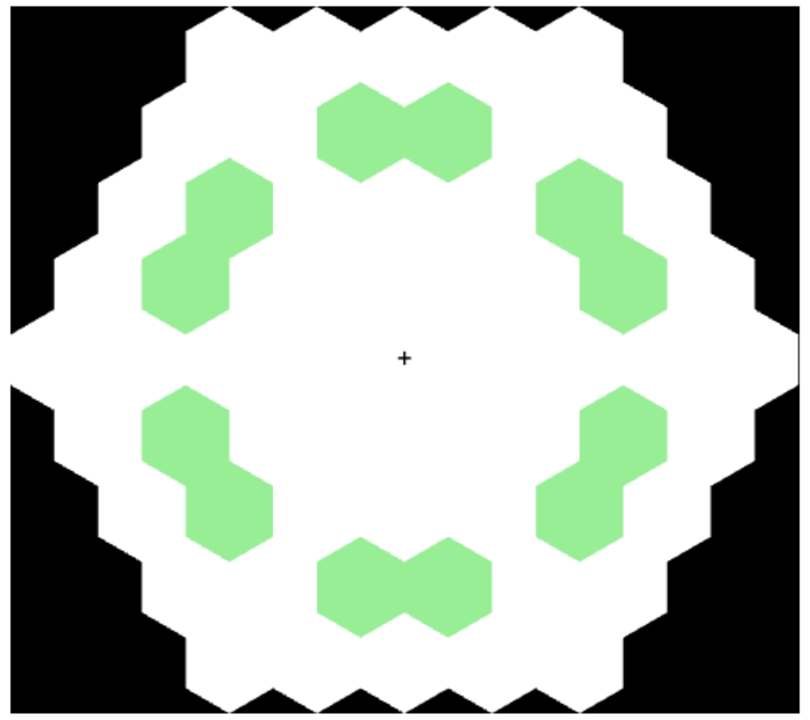
Reflect**O**rient**R**eflect**O**rient**R**eflect



cbdrwr

Corner**B**ar**D**elete**R**eflect...

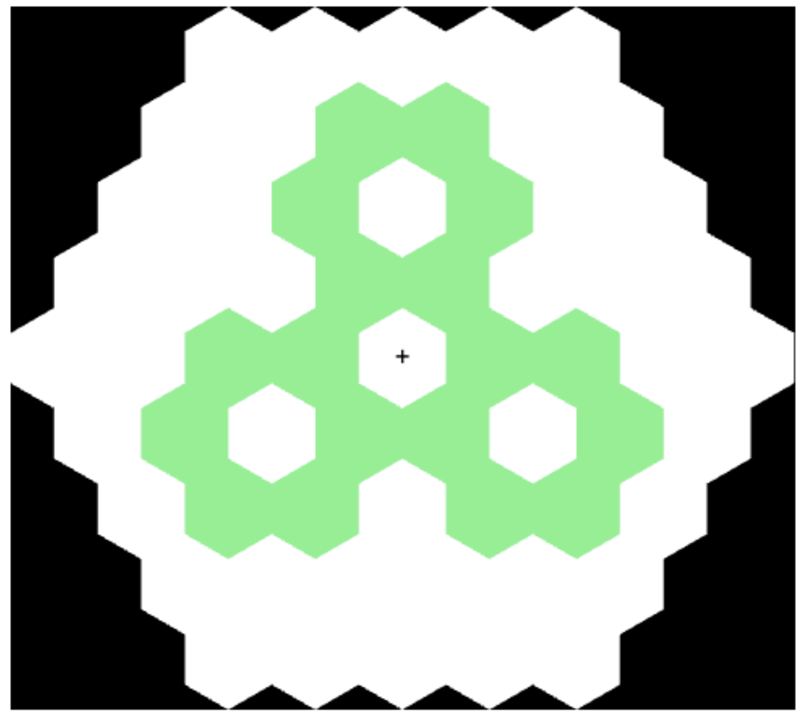
West**R**eflect



cdssroror

Corner**D**elete**S**outh**S**outh...

Reflect**O**rient**R**eflect**O**rient**R**eflect

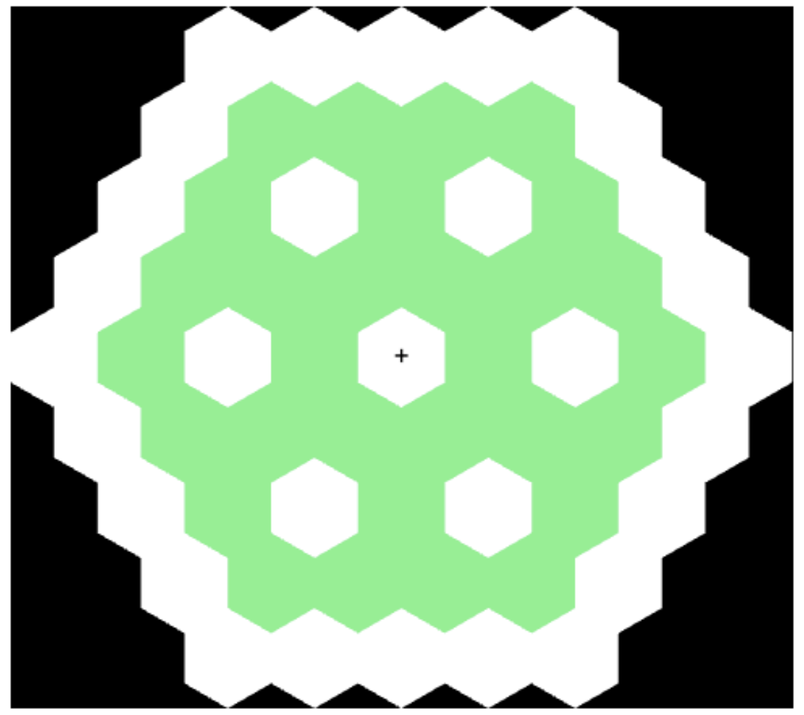


cbdreesror

Corner**B**ar**D**elete**R**eflect...

(north)**E**ast(north)**E**ast**S**outh...

Reflect**O**rient**R**eflect



cbdrwwroror

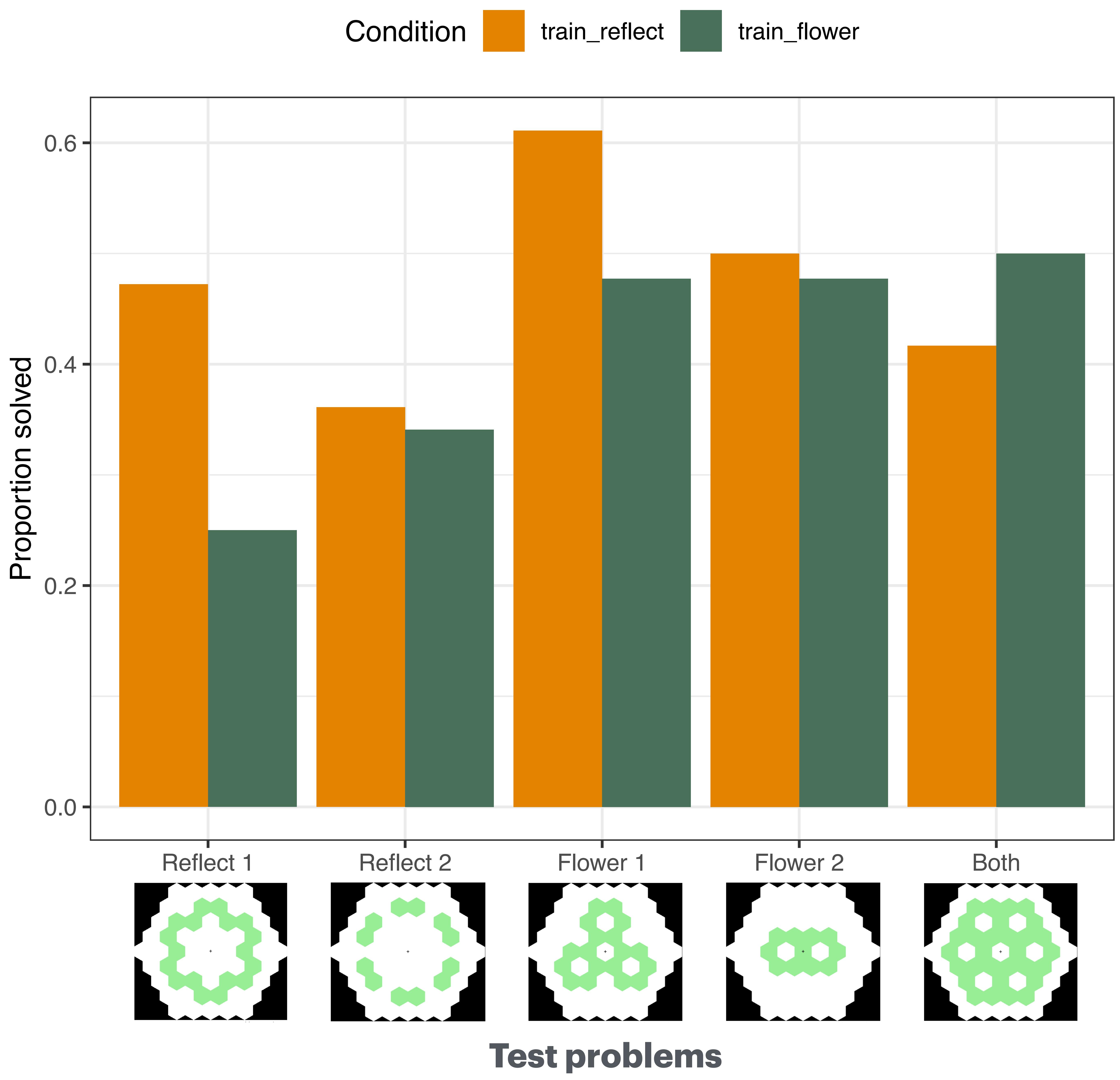
Corner**B**ar**D**elete**R**eflect...

West**W**est...

Reflect**O**rient**R**eflect**O**rient**R**eflect

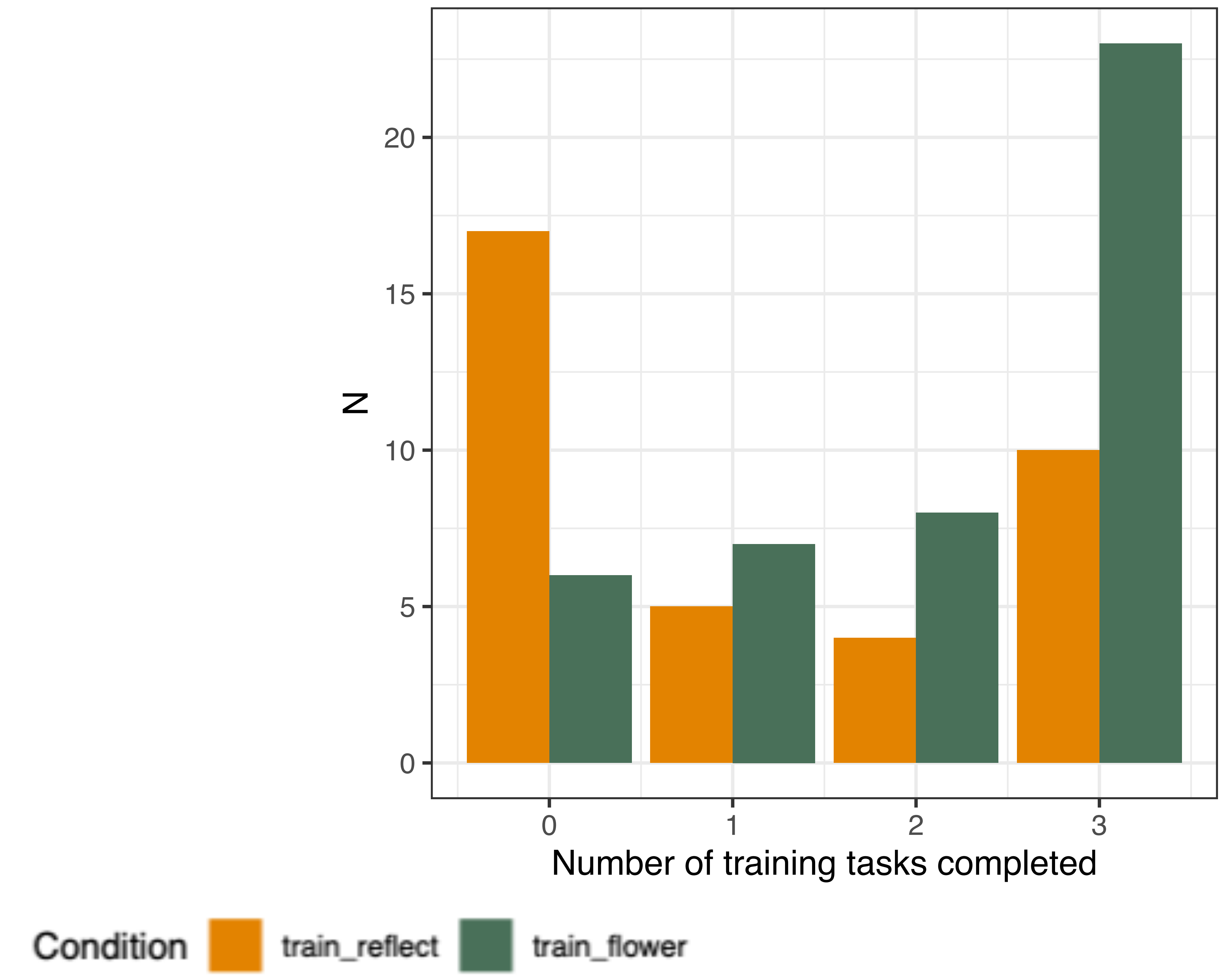
Performance

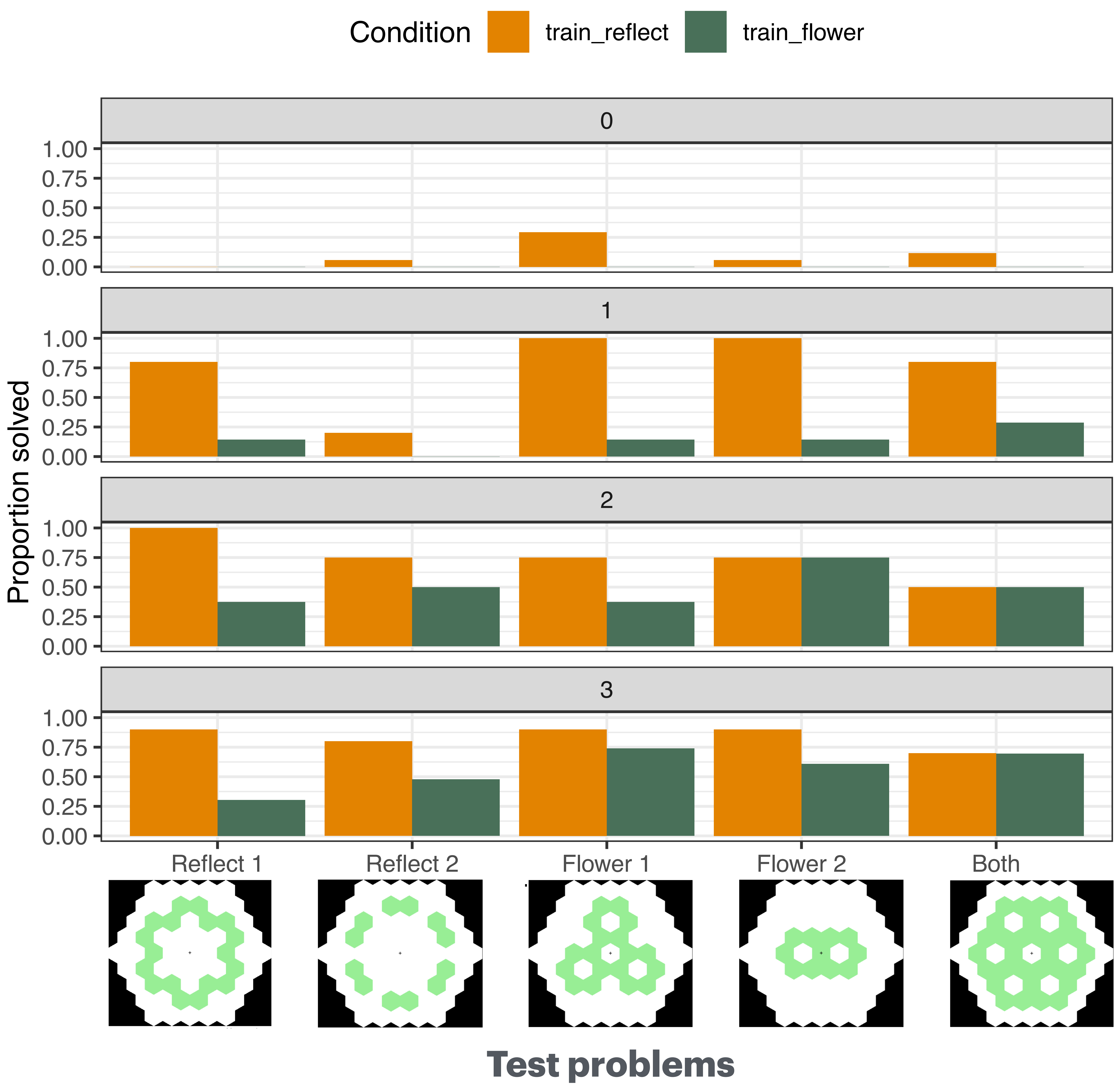
- Task is hard: **44**% / **51**% of puzzles solved (12% of all attempts)
- Mixed-effects logistic regression predicting p(Solve):
    - No main effect of Condition
    - Worse at Reflect 1 (p=.044) and Reflect 2 (p=.044), taking "Both" as baseline
    - Flower trained participants then worse at Reflect 1 test problem (p=.007), *but also marginally worse at Flower 1* (p=0.050) 🤷‍♂️



Performance

- Reflect training was harder
- Participants that succeeded in Reflect training did better at test in all problems





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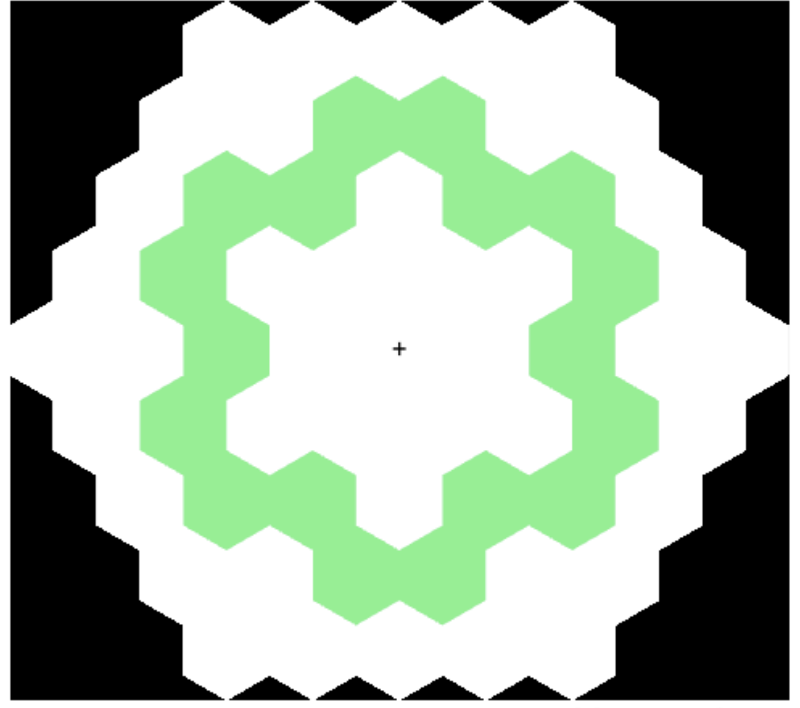

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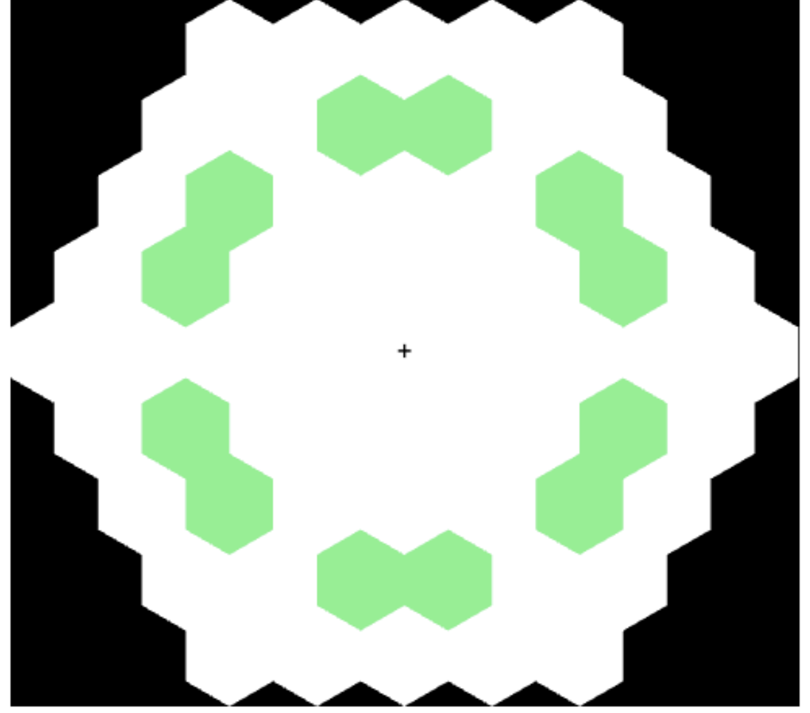
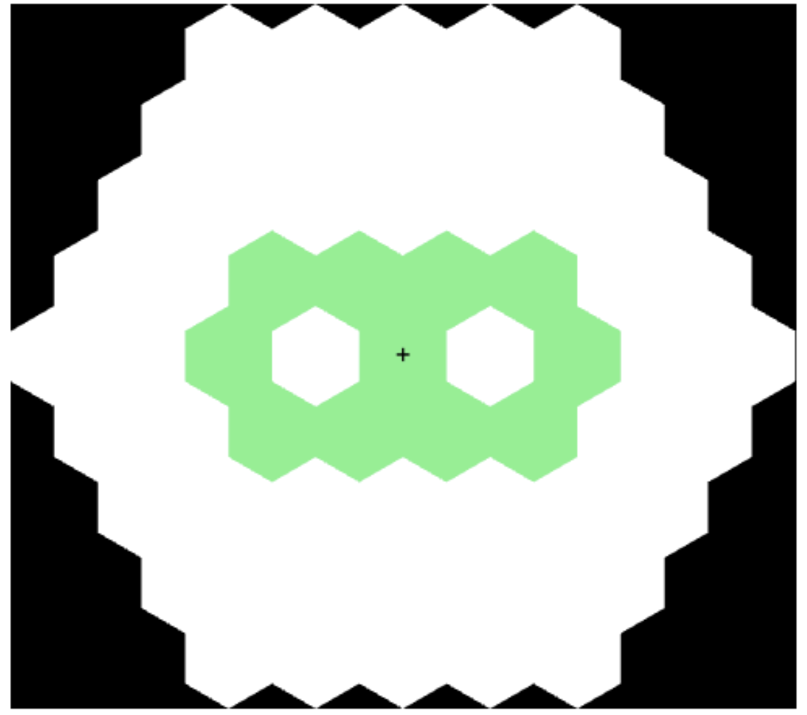


A great diversity of solutions were found!

Condition train_reflect train_flower



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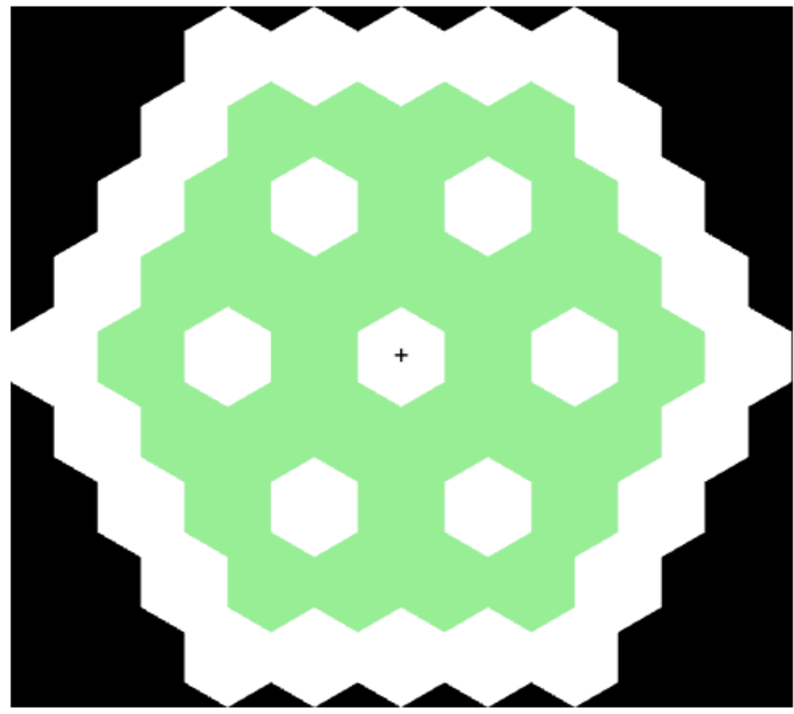

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Condition train_reflect train_flower

Extremely preliminary MCMC analysis

- Inferring individual participants libraries:
- Likelihood: The sum of the random-generation probability, of all possible parses of the action data with a candidate library
- Prior (favouring simpler grammars)
- Proposal:
  - Either: concatenate existing elements and add/upweight
    Or: decrement downweight on existing (nonbasic) element
    (Or: Introduce/delete a non-terminal)
- Acceptance function: Metropolis-Hastings (with detailed balance correction): stochastic acceptance based on unnormalised prior*likelihood (correcting for probability of proposing in either direction L|L′ / L′|L)

{A→AA,
A→**a**, A→**b**, A→**c**
A→**ab**}

{A→AA,
A→**a**, A→**b**, A→**c**,
A→**ab**}

{A→AA,
A→**a**, A→**b**, A→**c**
A→**ab**,
A→**abc**}

Condition train_reflect train_flower

Participant libraries

Welcome to the library of Babel

ww
oo; ro
oo; ww
br; ee
ss; ro
ss; ee
oo; or
ww; ror
cw; ooo
cb; asa
cc; cccc
ew; cw; ro
ee; ro; ww
se; ww; ee
or; rr; cbr
ro; coo; ooo
ss; rf; ro; we

ww; or; es; ew
fa; ad; bc; afa
ro; ww; oo; ror
we; ro; roo; ror
ss; oo; ee; ww; we
oo; ee; rr; or; ww
ss; ew; es; we; ww
oo; er; ww; ro; we; ss
ooo; cd; dd; bd; ew; ss
as; ww; dd; aa; aee; sa
ww; ro; oo; ss; we; ooo
ro; ww; aw; ee; ew; aa; as
ro; es; ee; wa; ww; as; ae
as; ww; we; ew; ss; ee; es; sw
cc; ss; as; sw; ee; bb; ac; we
sw; sa; wae; ea; wa; ss; ee; ww; es

we
es
ob; ro
we; oo
we; or
oo; cd
ew; oo
ro; aw
ra; ww
ooo; or
oo; ooo
oo; eee
ew; cooss
ro; cooooo
oo; cd; we
wr; ff; oo
we; oo; ee
br; cr; ww
we; cc; cww

ee; dsa; oo
acb; rd; we
or; oo; ff; rr
bobob; coc; oo
oo; es; ww; ssc
ee; oo; coo; ww
oo; cooo; we; ew
oo; cooo; ro; sc
we; ws; ro; es; oo
oo; bc; ae; cb; aw
ww; cb; oo; ee; ss
ww; ee; oo; ff; ss
ww; es; we; or; sw
ew; ae; as; cd; ww
oo; ae; ww; ee; ss
ae; ss; ww; ee; ea
ww; oo; aw; aa; awa
we; es; cwec; ws; dwe
or; oo; se; acw; ob; ror
ooo; we; cw; es; ror; ew
ww; we; ff; oo; rr; ee; dd
oo; co; bobo; ooo; bd; do; bdo; bobobd

How do people feel about the task?

"It was very challenging but fun"

“I enjoyed the task but found it frustrating at the same time I was pleased when I solved some but did find it very hard generally"

"It was very challenging but fun"

"I found it quite difficult but it was fun too"

"I liked it lots of symmetry involved"

"That was completely impossible I have never done a study that caused me to throw stuff across the room until now I am great at puzzles much better then my friends and coworkers I am always finding new puzzles to do"

"Ridicules and very frustrating Didnt enjoy at all and just wanted to finish it and get it over with"